Complement System

Lecture outline

- a. Describe the classical pathway of complement activation in terms of C1 and its subunits; C4 and its subunits; C2 and its subunits; and C3 and its subunits.
- b. For the alternative complement pathway, list the activating substances for initiation of this system and describe the role of C3b, factor B, factor D, properdin, factor H, and factor I in this pathway.
- c. Describe the three different functions—recognition, activation, and membrane attack.
- d. Discus the synthesization and composition of complement.
- e. Describe regulators of the complement system, both fluid phase and membrane bound.

procedure: bacteria added with human/mouse serum with antibody against staphylococcus → lysis
when they heated the antibody → no lysis
why? antibody is heat sensitive (complement system)

Contents

Discovery

Bac + Fresh serum (specific Abs)→ Lysis of Bac

Bac + heated serum at 56^{0}C (specific Abs) → No lysis

* complement system can be killed by heating

General properties

1. comprise more than 30 circulating and mem-expressed proteins
2. Most of them are proteases
3. Synthesized in the liver and by cells involved in the inflammatory response.
3. Heat-labile killed at 56⁰C for 30 min
4. Normally they are inactive and become activated due to microbes (presence of antigen)
5. Do not increase upon immunization
6. One of the major effector mechanism of humoral immunity and an important mechanism of innate immunity
7. Their serum conc. in human vary between 20 µg/ml of C2 and 1300 µg/ml of C3

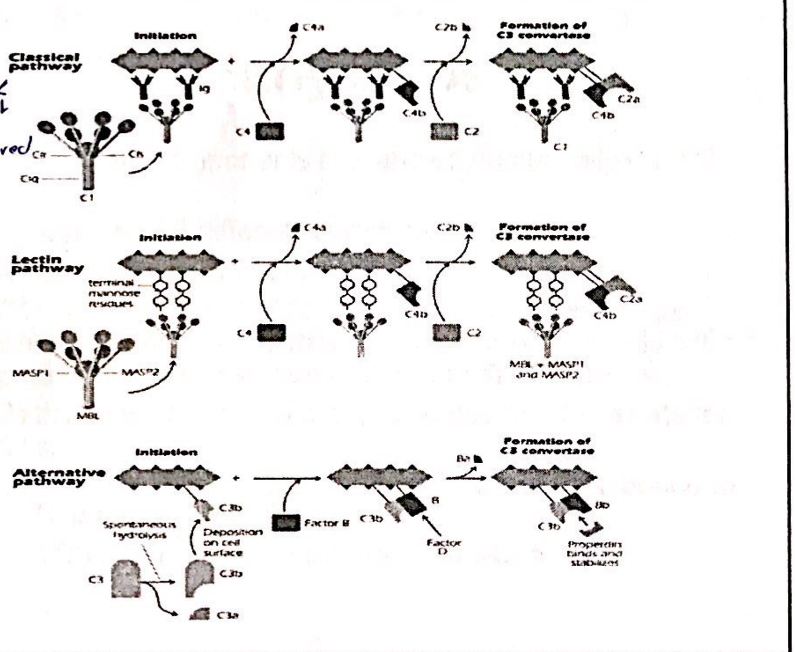
complement 2

* if we want to seperate the complement system we use C3

* complement system has 3 pathways:
1. classical pathway
2. Lectin pathway
3. Alternative pathway

* at first the activation in the 3 pathways is different but then they become the same pathway

the first to be discovered



* C1 consist of 3 enzymes:
1- C1Q
2- C1R
3- C1S

* activation of C1Q will activate C1R then C1S
* now that 3 enzymes are activated they need a substrate which is C4
* C4 breaks into two parts:
1- C4a the small part
2- C4b the large part

Classical Pathway

Activators C1Q bind with Fc region

- Ag-Ab complexes: C1 binds Fc region of 2 closed IgG or 1 IgM molecule
- Aggregated Ig
- C-Reactive protein (CRP) that bind surface phosphocholine of bac (CRP-PCh)
- Some viruses HIV

Not all classes of Ig are equally effective:

IgM> IgG3>IgG1>IgG2

- T-dep and TI Ags elicit IgM i.r and both activate Complement system

* once the Ag-Ab reaction happens, a change in tertiary structure happens and change in conformation that allows this binding thus free antigens don't bind with C1Q

* C4b will bind to the antigen
* C2 ~~and~~ is also a substrate
* C2a is the large part and C2b is the small part
* Ag-Ab-C1-C4b-C2a → a new enzyme called C3 convertase
* the substrate of C3 convertase is C3
* IgG and IgM activate the complement system
* IgM activate it more than IgG

Early steps

- The protein components are labeled numerically C1, C2 itc……
- C1 consist of 3 different enzymes: q, r, s
- C1q binds Fc region of Ig-bound to Ag
- C1r become activated followed by C1s C1 esterase: cleave C4 into C4a, C4b. C4a small remains in fluid → blood phase and C4b linked **covalent** to surface pathogen
- **Pathogen-C4b** then binds C2 and become Substrate for C1s.
- C1s cleave bound C2 into C2b in fluid and C2a bound to C4b **pathogen-C4b-C2a**

C3 convertase = pathogen-C4b-C2a

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Lectin Pathway

* by definition Lectin means protein recognize sugar

Activators

Teminal mannose expressed by (Lectin bind mannose on the surface of bacteria) → activation step

Gm+ Bac: *S aureus, H influenzae*,

Gm- Bac: *Klebsilla, E. coli*

Fungi: *Candida* , *Aspergillus*

Mannose is not found on the surface of mammalian cells—self and nonself discrimination → mannose does Ab job with determining which is self and nonself

- **Early steps**

mannose binding Lectin

-Pathogen-mannose bind (MBL) protein that is homologous to C1q in the classical pathway bind mannose-associated serine proteases (MASP). MBL-MASP-1 and MASP-2

-Pathogen-mannose—MBL—MASP-1—MASP-2

-MASP-2 cleave bound C2 into C2b in fluid and C2a bound to C4b **pathogen-C4b-C2a**

C3 convertase = pathogen-C4b-C2a

Alterative pathway

Activators

- **Any foreign substance in the absence of Ab**

Early steps

very small fraction of C3 is hydrolysed

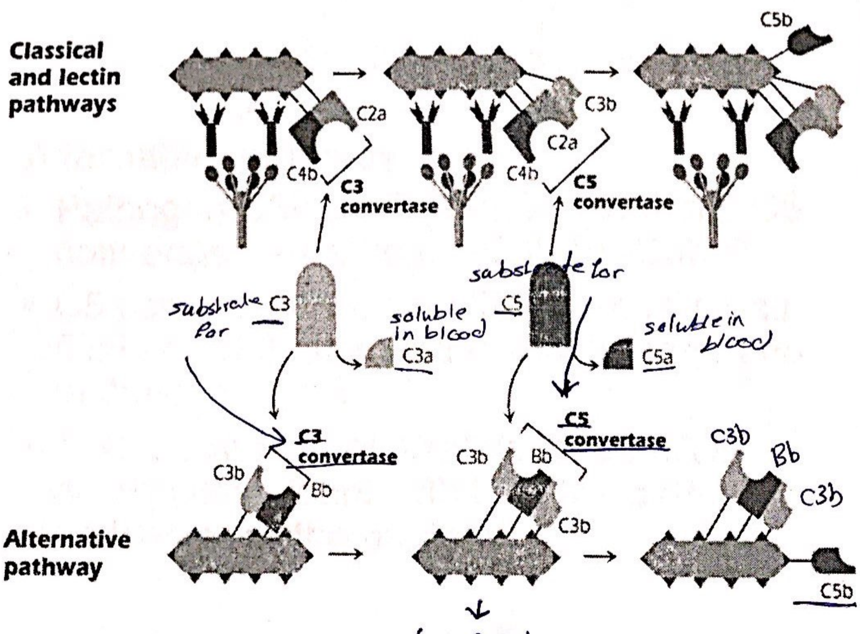
- C3 spontaneous hydrolysis to small fragment C3a and large C3b cleavage of reactive thiol group
- C3b deposition on the cell surface of pathogen or mammalian (bind to proteins and CHO on surfaces).

AP is always ON. Later Regulatory steps

- Pathogen-C3b bind to another protein Factor B; Pathogen-C3b-B

becomes a substrate called factor D

- Factor D split B into small Ba and large Bb fragment—(Pathogen-C3b-Bb-Properdin factor) C3b-Bb is unstable so to stabilize the complex Properdin bind



once C5b bind on the surface of the bacteria other complement proteins follow in sequence (C6, C7, C8, C9)

the most important, is a monomers, becomes a polymer by polymerization

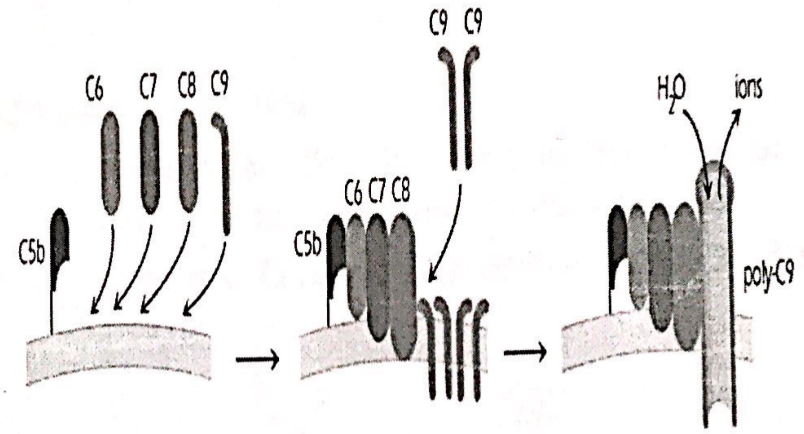
Activation of C3 and C5

Classical and Lectin pathways

- C3 convertase cleave C3 into small fragment C3a and large C3b
- C3b binds C4b-C2a to make Pathogen-C4b-C2a-C3b (C5 convertase)
- C5 convertase cleave C5 into small C5a in fluid and C5b attached to the pathogen not to the convertase

Alternative pathway

- Pathogen-C3b-Bb-P bind C3b to make C5 convertase ---Pathogen-C3b-Bb-C3b-P
- C5 convertase cleave C5 into small C5a in fluid and C5b attached to the pathogen not to the convertase
- This is example of positive feedback or amplification loop: C3b bind the pathogen and increase the process



Terminal Pathway

- Pathogen-C5b will bind to C6-C7-C8 so C5b-C6-C7-C8 is a Receptor for C9
- C9 A perforin-lke component binds to C8. Polymerization of C9 monomers occurs to form a transmembrane structure called MAC or **membrane attack complex** → lysis of bacteria
- The MAC will disturb the osmotic equilibrium of the pathogen and the membrane becomes permeable to macromolecules and cell lysis

Regulation of complement system

C1 esterase INH

- Serum protein that inhibit the first step of classical, lectin and alternative pathway
- C1INH bind C1s and MASP-2 and C3bBb

Inhibition of C3 convertase

4 inhibitors for 3 pathways

Classical pathway

- (1) C4b-binding protein (C4bBP), (2) Decay accelerating factor (DAF), CR-1 (4) and Membrane cofactor protein (MCP) (3).
- These displace the C2a so C4b-CR-1, C4b-BP, MCP This make good substrate for factor I that cleave C4b into C4c and C4d
- DAF dissociate C4b-C2a but does not act as cofactor for Factor I

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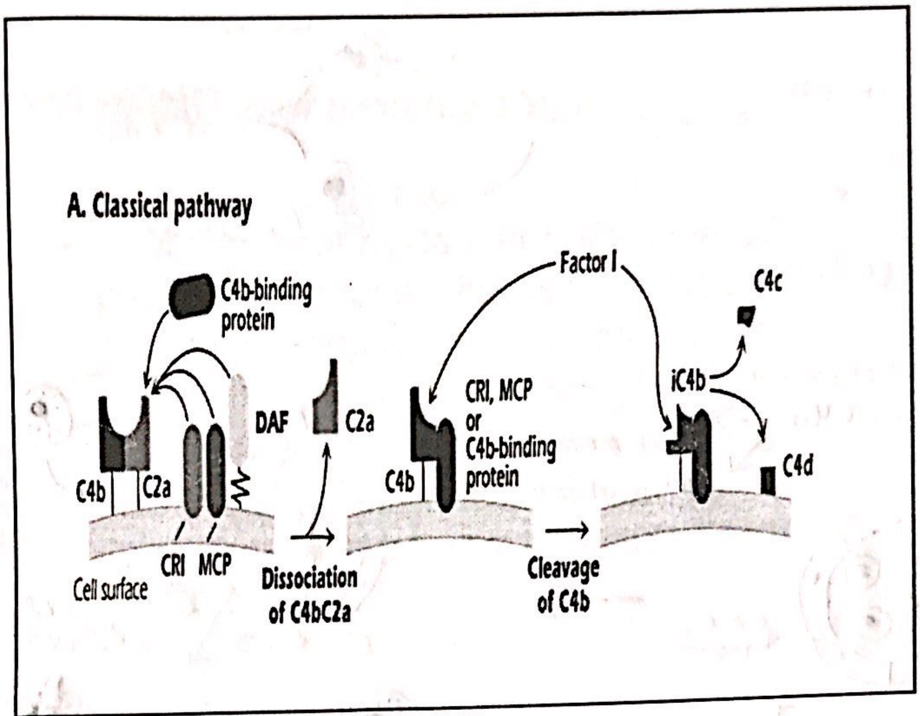
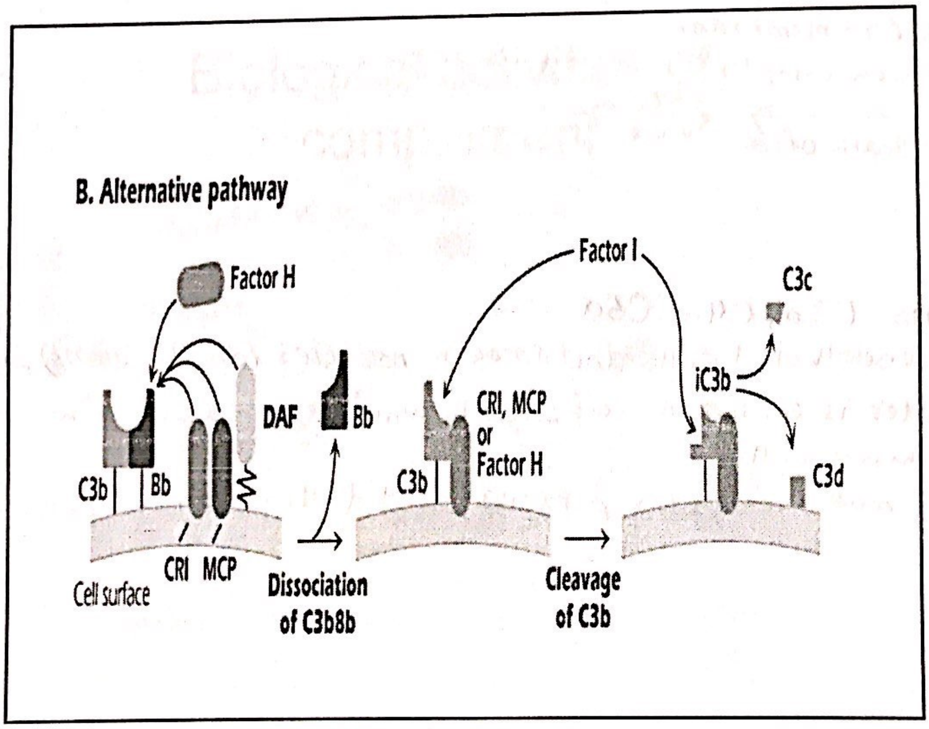
then C3 convertase becomes

- Factor H, DAF, CR-1 and MCP displace Bb in the alternative pathway
- Factor I cleave C3b to C3d and C3c.
- DAF dissociate C4b-C2a but does not act as cofactor for Factor I

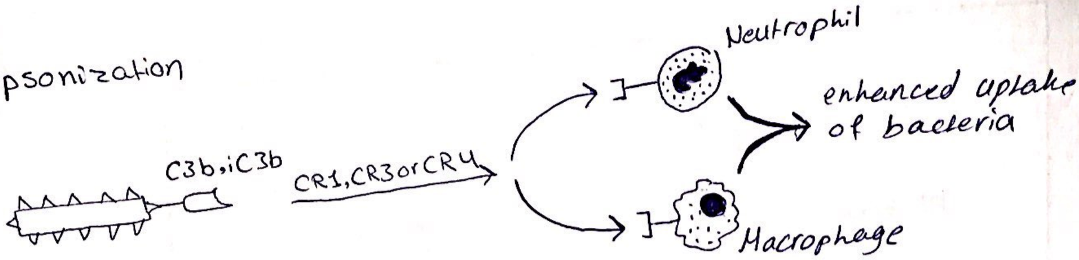
Alternative pathway

- Factor H: 2 functions in the alternative pathway:
- compete with factor B to bind C3b on cell surface
  Sialic acid found on mammalian cells favor factor H while Bacteria lack sialic acid favor binding C3b or B factor.
- binds to C3 convertase , displaces Bb, prevent further activation and become cofactor for I

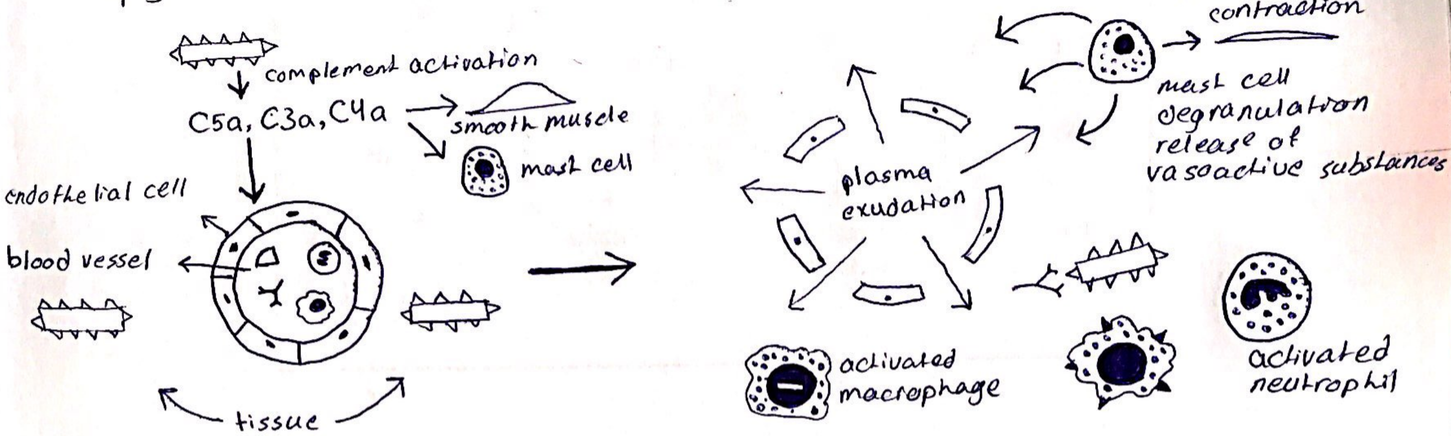

A. Classical pathway
C4b-binding protein
Factor I
C4c
iC4b
DAF
C2a
CRI, MCP or C4b-binding protein
C4b
C2a
C4b
C4d
CRI
MCP
Cell surface
Dissociation of C4bC2a
Cleavage of C4b


B. Alternative pathway
Factor H
Factor I
C3c
iC3b
DAF
Bb
CRI, MCP or Factor H
C3b
Bb
C3b
C3d
CRI
MCP
Cell surface
Dissociation of C3b8b
Cleavage of C3b

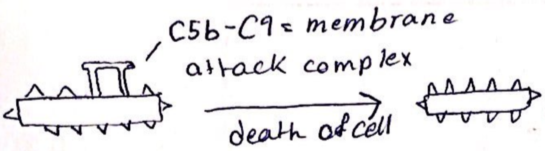
Opsonization



Anaphylatoxin production



Cell lysis



* anaphlatoxins : C3a, C4a, C5a
they work on endothelial cells and smooth muscles (on the viens), the function in smooth muscles is contraction of smooth muscles (لتوسع الشرايين لزيادة تدفق الدم عن تعرض الجسم لجسم غريب) and allergy in mast cells.
anaphlatoxins makes vascular permeability (retraction of cells)

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Regulation of terminal steps

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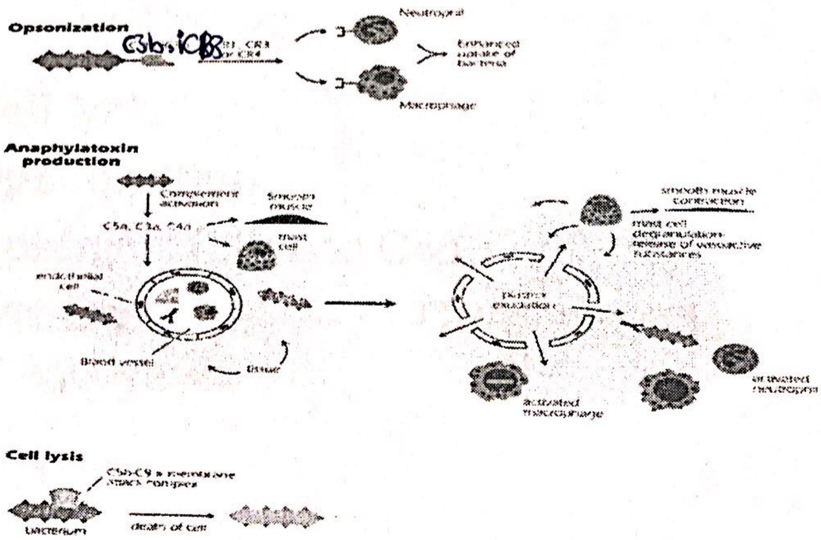
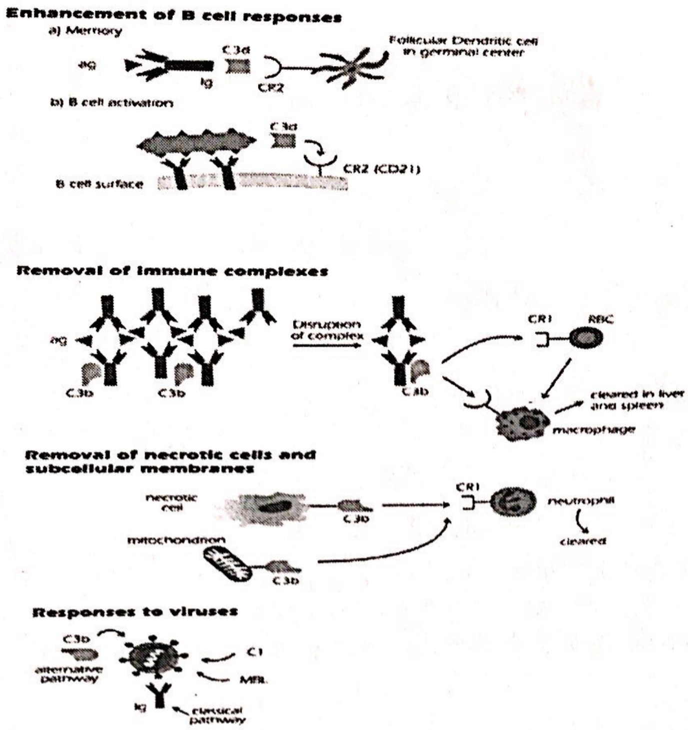
normal cells

- a. innocent bystander cells have CD59 protein on surfaces that prevent lysis of cells by binding to C5b-C8 complex so preventing C9 polymerization (MAC)
- b. Vitronectin and clusterin proteins fluid proteins bind C5b6 and C5b6,7 C5b6,7 ,8 and prevent interaction with membranes

Biological activities of complement





Biological activities of complement

- **Cell lysis**
- **Opsonization**

  Opsonins: C3b and C4b

  Bac-Opsonin-------------CR-phagocyte phagocytosis

Anaphlatoxins

- C5a> C3a> C4a potency
- Bind receptors on blood vessels endothelial cells:
  a.increase V. permeability---edema ---immune Abs and cells to enhance i.r
  b. Chemotactic for neutrophils
  c. induce smooth muscle contraction
  d. interaction with mast cells and basophils release histamine and other mediators that increase v. permeability and smooth muscle contraction

*C3b (opsonization) bind C3b with Ag and makes it succebtible to phagocytosis
- removal of immune ~~response~~ complex
- removal of necrotic cells or dead parts of the cell

Enhancement of B cell response

a. Activation of B cells:

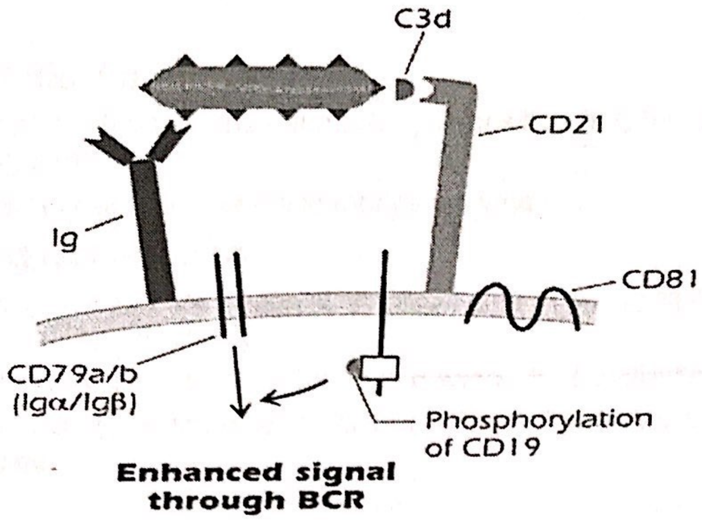
C3d bind CR2 (CD21) on B cells and enhance i.r 1000 folds bind Ag, enhance B cell responce

Surface B cell—sIgM—Ag—c3dg—CD21—surface B cells

b. Induction of memory cells

depletion of complement by copra venum factor interferes with memory cell formation and secondary i.r.

(Experiment)



Removal of Immune complexes

- Ag + multivalent Ag like bac → large insoluble IC
  In vitro: usefull in diagnisis
  In vivo: can be detrimental to the host
- Deposition of C3b breaks up the large IC into smaller pieces----cleared by phagocytosis
- Deposition of C3b laso allows binding RBCs with CR1---circulation to liver and spleen---phagocytosis

Removal of necrotic cells

Physiological process

- Dying cells activate complement S. by deposition of C3b---followed by phagocytosis
- Subcellular fractions like mitochondria activate c

Pathological process

Ischemia: area of tissue dies after blood and O2 supplies have been cut---cardiac attack

Reperfusion: attempt to restore blood supply to the affected tissue.

C is considered a major contributor to the inflammatory response

Response to Viruses

- C plays a role in defense against viral infection including retroviruses, lentiviruses
- Escape mechanisms
  - Some V produce proteins that mimic C inhibitor like DAF
  - Some V bind to CR such as EBV bind CR2
  - Some V bind C3b, bind the host CR. HIV use CR2 to infect T, B cells

Complement Deficiencies

- C3 def.
- C3 a and C3b are not produced
  - Terminal components are not activated
    - severe pyogenic (pus-forming) bacteria infections including: N meningitidis, S pneumonia, H influenzae, S aureus